

Guide for Using the Chart of the Nuclides

Chemical Element

H – Symbol

1.00794 – Atomic Weight (Carbon-12 Scale)

Hydrogen – Element Name

σ_a .333, .150 – Thermal Neutron Absorption Cross-Section in Barns Followed by Resonance Integral, in Barns

Stable – Even Z, Even N Nuclides Have Spin and Parity 0⁺

Pd108 – Symbol, Mass Number

26.46 – Atom Percent Abundance

Thermal Neutron Capture Cross-Sections in Barns Leading to (Isomeric + Ground State), Followed by Resonance Integrals Leading to (Isomeric + Ground State).

σ_γ (.19+8), (2+24E1)

107.903895 – Isotopic Mass (Carbon-12 Scale)

Artificially Radioactive

S38 – Symbol, Mass Number

2.84 h – Half-Life

Modes of Decay with Energy of Radiation in MeV for Alpha and Beta; keV for Gammas.

β^- .99, ...

γ 1941.9, ...

E 2.94 – Beta Disintegration Energy in MeV

Displacement Caused by Nuclear Bombardment Reactions

α , 3n	α , 2n ${}^3\text{He}$, n	α , n	
p, n	p, γ d, n ${}^3\text{He}$, np	α , np t, n ${}^3\text{He}$, p	
p, pn γ , n n, 2n	Original Nucleus n, n	d, p n, γ t, np	t, p
p, α	n, t γ , np n, nd	n, d γ , p n, np	n, p t, ${}^3\text{He}$
	n, α n, n ${}^3\text{He}$	n, ${}^3\text{He}$ n, pd	

Naturally Occurring or Otherwise Available but Radioactive

Symbol, Mass Number – **La138** 5+

– Spin and Parity

0.090 – Atom Percent Abundance

1.05E11a – Half-Life

Modes of Decay in Order of Prominence with Energy of Radiation in MeV for Alpha and Beta; keV for Gammas.

ϵ , β^- .25

γ 1435.8, 788.7

σ_γ ~57.4E2

E 1.04 137.90711 – Beta Disintegration Energy Followed by Isotopic Mass

Thermal Neutron Capture Cross-Section, Followed by Resonance Integral.

Member of Naturally Radioactive Decay Chain

Historical Symbol – **Po218** **RaA** 3.10 m

– Symbol, Mass Number

– Half-Life

α 6.0024

γ 510.

β^- ω

ω Indicates Decay Mode Intensity (See Symbol List)

218.008965 – Isotopic Mass

Two Isomeric States One Stable

Spin and Parity of Metastable State, 11/2⁻

Half-Life – **11/- Sn117** 1/+

– Spin and Parity of Ground State, 1/2⁺

– Symbol, Mass Number

– Atom Percent Abundance

Modes of Decay with Energy of Radiation in keV.

IT 156.0, e^-

γ 158.6

σ_γ 1.1, ~15

Thermal Neutron Capture Cross-Section in Barns, Followed by Resonance Integral in Barns

116.902955 – Isotopic Mass

– Fission Product, Slow Neutron Fission of U235, U233 or Pu239.

– Stable Ground State

Two Isomeric States Both Radioactive

Spin and Parity of Metastable State – **2+ Co60** 5+

– Spin and Parity of Ground State

– Symbol, Mass Number

– Half-Life

– Atom Percent Abundance

Modes of Decay and Energy in Order of Intensity; ... Indicates (Where Shown) Range of Energies Included.

IT 58.6, e^- β^- .318, β^- 1.6 ω ...

γ 1332.5 γ 1332.5, 1173.2, ω ...

σ_γ 6E1, 2.3E2 σ_γ 2.0, 4 E 2.824

Thermal Neutron Capture Cross-Section in Barns Followed by Resonance Integral in Barns

– Beta Disintegration Energy in MeV

Radioactive m-State Isomer

Radioactive g.s. Isomer

Relative Locations of the Products of Various Nuclear Process

		${}^3\text{He}$ in	α in
β^- out	p in	d in	t in
n out	Original Nucleus	n in	
t out	d out	p out	β^+ out ϵ
α out	${}^3\text{He}$ out		

Radiation and Decay

α	alpha particle
β^-	negative electron (beta decay)
β^+	positron
γ	gamma ray
n	neutron
p	proton
d	deuteron
t	triton
ε	electron capture
IT	isomeric transition
D	delayed radiation (see page 6)
SF	spontaneous fission
E	disintegration energy
e^-	conversion electron
$\beta^-\beta^-$	double beta decay
C14	particle emission (C14 nuclei)
Ne	particle emission (neon nuclei)

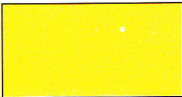
Time

μ s	microseconds (1.0E-6s)
ms	milliseconds (1.0E-3s)
s	seconds
m	minutes
h	hours
d	days
a	years

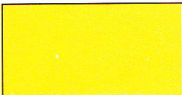
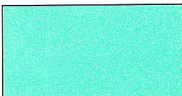
Other Symbols

ω	absolute abundance less than 1%
$\nu\omega$	absolute abundance less than 1.0E-3%
$\nu\nu\omega$	absolute abundance less than 1.0E-6%
E	indicates exponential format, e.g., 1.06E11a is 1.06×10^{11} years
$\longleftrightarrow$	indicates assignment to metastable (m) and ground (g) states inconclusive
$\updownarrow$	indicates assignment of m_2 and m_1 states inconclusive
χ	unspecified number of particles of a given type emitted, e.g., χp
	nuclide formed as a product in fission of U235, U233 or Pu239
	shielded nuclide with independent fission yield greater than 2.5E-6%

Colors Used for Half Lives (Appear in Upper Half of Nuclide Block)

	1 Day to 10 Days
	10 Days to 100 Days
	100 Days to 10 Years
	10 Years to 5E8 Years
	> 5E8 Years or Stable

Colors Used for Neutron Absorption Properties (Appear in Lower Half of Nuclide Block)

	10 Barns to 100 Barns
	100 Barns to 500 Barns
	500 Barns to 1000 Barns
	> 1000 Barns